Horizontal launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find flight time from vertical motion and then horizontal range.

SUCCESS CRITERIA - I can...

1. Separate vertical and horizontal givens.
2. Find or use fall time before calculating range.
3. Check units and interpret the landing point.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

At what altitude h/R is a spherical planet's field 1/9 of its surface value? Repeat for 1/25.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $h = \frac{1}{2}gt^2$; range = $v_x t$.
Horizontal launch; h is the positive drop, t is fall time; g = 10 m/s². CPA may use the supplied time.

Worked reasoning: A horizontal launch falls 20 m with g=10 m/s². From $20=5\ t^2$, t=2 s. At horizontal speed 3 m/s, range=6 m.

Your turn: Draw a diagram labeling height and range. Number the two solution steps to show why time comes first.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and interpret the model

On the diagram, label height h, horizontal range and initial horizontal velocity. Which direction determines the fall time?



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

In a launch simulation, a ball leaves a 45 m cliff horizontally at 15 m/s. Label height and range on the model. Find the flight time first, then the range. Use g = 10 m/s² and ignore air resistance.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

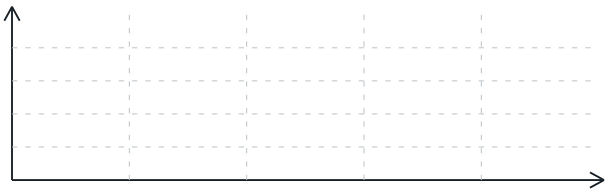
Two platforms have heights 5 and 20 m. Supplied fall times are 1 and 2 s. At the same launch speed, does quadrupling height quadruple range? Explain.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simplified model predicts position (-60, 80) km with bounds ±6 km in x and ±8 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the model.

y position (km)



x position (km)

Marzano check - circle: 1 2 3

07 Exit check

From a 20 m cliff, horizontal launch speed is 8 m/s. Find t, range and final vertical velocity. Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Separate horizontal and vertical motion

Label x and y; state positive directions.



Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A horizontal launch at 12 m/s must land 18 m from the base. Find launch height and impact speed using g=10 m/s². Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3